Natural History

Alveary

Alveary

Science

Natural History: Grade 6

GRADE(S): 6

About the Course

Students learn about oceanography, predator-prey relationships, and the impact of pollution as they consider man's stewardship of the natural world. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 6

About Science: Grade 6
In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level and time required are transitional as we gently prepare for the middle grades.

Science: Grade 6
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more "friends" in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge "by the way" and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 6
In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level, historical context, and time required are transitional as we gently prepare for the middle grades.

Natural History: Grade 6
For approximately sixth-grade students based on a balance of complexity and reading

level. However, learners may be freely combined, or sequence reordered based on needs and preferences.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
6	Natural History: Grade 6 1 time/week 15 min	The Frog Scientist The Wolves and Moose of Isle Royale: Restoring an Island Ecosystem Tracking Trash

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 6				
Natural History: Grade 6		General Science: Grade 6 Nature Walks & Scouting: Grades 1-8	General Science: Grade 6 Nature Notebook: Grade 6	Labs: Grade 6

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 6
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 6
Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

Natural History: Grade 6
☐ Term 1: shore, stream, or reservoir, a marine animal rescue
☐ Term 2: wildlife conservationist
☐ Term 3: a pond (hopefully one where tadpoles may be collected)

Term Prep & Teacher Tips

Science: Grade 6
Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term

and/or any that support special topics chosen by students:

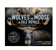
Natural History: Grade 6
☐ Term 1: -
☐ Term 2: -
☐ Term 3: amphibians p.170-187

Books & Resources

For book rationales and purchase options, click the Book List link or
scan the QR code below.
∞ View Book List Details

Natural History: Grade 6

 The Frog Scientist

 The Wolves and Moose of Isle Royale: Restoring an Island Ecosystem

 Tracking Trash

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or
scan the QR code below.
∞ View Basic Supplies
(No Subject Supplies Assigned)

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

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Natural History: Grade 6

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

• These can be viewed as alternative ways to engage.

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Term 1

WEEK 1

⬚ **15m Natural History: Grade 6 - Lesson 1**
Ocean Currents

Materials: Tracking Trash

PREP: Read Teacher Tip.

→ INTRO
Look at the picture on the cover and the title. Have you ever noticed trash on the beach or in a stream? This book is about how one scientist uses that trash as an opportunity to learn more about the ocean and the creatures that live within it. Examine the map opposite p.1 and notice the current. Why do you suppose the mapmaker was interested in that current, and what does that have to do with our story?

→ READ, NARRATE, & DISCUSS
p.1-4 "Benjamin Franklin" - "learn from it."

• How do these ocean currents affect the climate? You can study the map on p.20-21 of your NatGeo World Atlas.
• Explain weather versus climate.

★ TEACHER TIP
If students read Tornado Scientist, they might recall that air masses are warmed by the Earth or water beneath them. Movement occurs based on their temperature and density, and this is a driving force behind weather patterns.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

⬚ **15m Natural History: Grade 6 - Lesson 2**
An Opportunity

Materials: Tracking Trash

PREP: Read Teacher Tip.

→ RECAP
Recall what this scientist is studying and why. Let's find out how he started tracking trash.

→ READ & NARRATE
p.4-7 "Curt didn't" - "Pacific Ocean."

→ READ, NARRATE, & DISCUSS
p.8-9 "Finding a certain" - "of the spill."

★ TEACHER TIP
This passage has two parts with two different purposes. The first part tells how this experiment began and is more likely to be narrated as a series of events. The second explains latitude and longitude (and why they are important) and is more likely to be narrated as points that describe these concepts. It can be helpful to teach students to identify the purpose of different passages as they prepare to narrate.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

⬚ **15m Natural History: Grade 6 - Lesson 3**
Drift Experiments

Materials: Tracking Trash

→ RECAP
Remind me how Curt started tracking trash. Scientists have known for a long time that ocean motion isn't random. How do you think scientists typically study ocean currents?

→ READ, NARRATE, & DISCUSS
p.11-15 "Throughout this" - "floating sneakers."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ **15m Natural History: Grade 6 - Lesson 4**
Modeling Ocean Motion

Materials: Tracking Trash

PREP: Read Teacher Tip.

→ RECAP
How were the scientists studying the ocean in the last passage? Today, we'll find out what happened when they used the OSCURS program to study the sneaker spill. We'll also learn about different types of ocean motion.

→ READ & NARRATE
p.15-17 "In 1992" - "more recoveries."

→ READ, NARRATE, & DISCUSS
p.18-19 "Imagine you" - "Indian Ocean."

→ SUPPLEMENTAL
∞ Video Link: How Do Ocean Currents Work? (4:34)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, liven up the lesson, or add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 1

WEEK 5

⬚ 15m Natural History: Grade 6 - Lesson 5

A Variation

Materials: Tracking Trash

PREP: Read Teacher Tip

→ RECAP
What did the scientists learn from the OSCURS computer program? What other kinds of objects do you think would make a good study? Look at the picture on p.20.

→ READ, NARRATE, & DISCUSS
p.21-26 "While they" - "brand of oceanography."

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in the ocean or ocean currents? More interest/ease in thinking about how the ocean impacts climate or wildlife? If not, consider observing together more often, exploring extra helpings, or taking a field trip to the ocean or waterway.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

⬚ 15m Natural History: Grade 6 - Lesson 6

Riding the Currents

Materials: Tracking Trash

→ RECAP
Recall what happened in the last passage. These ocean scientists continue to work with citizen scientists to make some interesting discoveries. So, we'll read about that and some microscopic ocean creatures that depend on ocean motion today.

→ READ & NARRATE
p.26-28 "It is at" - "to track it."

→ READ, NARRATE, & DISCUSS
p.29-31 "Sneakers" - "all of us."

→ SUPPLEMENTAL
∞ Video Link: The Story of Plankton (3:01)
∞ Image Link: What is Upwelling?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ 15m Natural History: Grade 6 - Lesson 7

Where Do Floating Objects Go?

Materials: Tracking Trash

→ RECAP
Recall what we read last week. They seem to use the OSCURS program a lot. How do you think a computer simulation is helpful if it's just a hypothetical experiment?

→ READ
p.33 "What would happen" - "might look like."

→ VIEW
Examine the figure on p.32, following the parts of the figure using the caption. If students want to look back at the North Pacific Subtropical Gyre, refer to p.19.

→ READ, NARRATE, & DISCUSS
p.34-37 "At the beginning" - "Garbage Patch."

→ SUPPLEMENTAL
∞ Image Link: Garbage Patch Photos

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ 15m Natural History: Grade 6 - Lesson 8

The Garbage Patch

Materials: Tracking Trash

→ RECAP
Tell me what they found out in the last passage. What do you think happens to everything in the Garbage Patch?

→ READ & NARRATE
p.37-39 "What happens" - "the problem."

→ READ, NARRATE, & DISCUSS
p.40 "People often" - "same thing!"

• Discuss the chain of cause and effect that begins with trash in the environment.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 1

WEEK 9	⬚ **15m Natural History: Grade 6 - Lesson 9** *Ghost Nets* ▯ Materials: Tracking Trash **→ RECAP** What did we learn in the last reading? When cleaning your room, have you ever found unexpected stuff that you didn't even know you'd lost? That's similar to what happens in today's reading. **→ READ, NARRATE, & DISCUSS** p.43-46 "In 1991" - "drift in them." • How do you think knowledge about ocean currents can help?	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 10	⬚ **15m Natural History: Grade 6 - Lesson 10** *In Search of Garbage* ▯ Materials: Tracking Trash **→ RECAP** Tell me about the discovery from the last reading. What do you think scientists are doing to solve the problem? What do you think the average person can do? **→ READ, NARRATE, & DISCUSS** p.46-51 "Our first" - "your mother!" • Interestingly, continued research has discovered that coastal creatures are beginning to colonize the Garbage Patch! You can read more about it in the link, as desired. Does the presence of life change how we think about the Garbage Patch? Should it? **→ SUPPLEMENTAL** ∞ Article Link: Garbage Patch Home	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 11	⬚ **15m Natural History: Grade 6 - Lesson 11** ▯ Materials: Tracking Trash PREP: Read Teacher Tip **→ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	★ TEACHER TIP When students take exams next week, evaluate what they recall, but more importantly, whether the ocean, its currents, or its connection to the ecosystem are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? **• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **15m Natural History: Grade 6 - Lesson 12** *Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: Tracking Trash	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 2

WEEK 1

⬚ 15m Natural History: Grade 6 - Lesson 13

Finding Isle Royale

Materials: The Wolves and Moose of Isle Royale

PREP: Read Teacher Tip.

➔ INTRO
Look at the picture on the cover. Have you seen or read anything about wolves or moose before? If so, what do you know about them? If not, what are your impressions?

➔ READ, NARRATE, & DISCUSS
p.1-6 "Rising out" - "and explore."

• Have you heard of ecosystems before? How would you describe an ecosystem? (Read the sidebar on p.2, as desired.) Look at the pictures on the page facing p.1 and on p.3. What do these photos tell you about the ecosystem featured in this book?
• Look at the map on p.4. Find Isle Royale on a map of North America. Did you know that there are moose on this tiny island? What else do you think might live there?

★ TEACHER TIP
We won't have time for many of the sidebars in this book, but you can share them with students, as desired.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

⬚ 15m Natural History: Grade 6 - Lesson 14

Arriving at Isle Royale

Materials: The Wolves and Moose of Isle Royale

PREP: Read Teacher Tip.

➔ RECAP
What did we read in the last passage? Where are we? Why are we here? Today, we'll begin the story of the place that is Isle Royale.

➔ READ, NARRATE, & DISCUSS
p.6-13 "The only way" - "their way there."

➔ SUPPLEMENTAL
∞ Image Link: History and Culture of Isle Royale

★ TEACHER NOTE
The Earth's age is mentioned, and the formation of the land is described on p.9. Teachers might take the opportunity to have a discussion or to share alternative theories.

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to liven the lesson, or to add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

⬚ 15m Natural History: Grade 6 - Lesson 15

Wildlife at Isle Royale

Materials: The Wolves and Moose of Isle Royale

PREP: Read Teacher Tip.

➔ RECAP
Recall what we learned in the last passage. Let's continue the story of Isle Royale. We'll find out how the moose and wolves connect to the ecosystem and what the author notes when she arrives there.

➔ READ, NARRATE, & DISCUSS
p.13-20 "When the earliest" - "research team."

• We don't have time to read all of the team bios, but look at the pictures of the Petersons on p.21. Lead scientist Rolf Peterson says Isle Royale is an "ideal natural laboratory" because humans have minimal impact here. Trees are not harvested; animals are not hunted or abused. However, Candy Peterson says that this perspective excludes the important role that humans have to play. We should be enjoying, appreciating, and taking responsibility for nature. Labs are for "tinkering," but she says that Isle Royale is more like a giant circus where the animals are in charge and we watch and observe. What do you think about that? As the story continues, think about these perspectives and how people with different perspectives can work together.

★ TEACHER TIP
In this passage, the first part introduces us to the dynamics of the ecosystem and the second to the author's first impressions of the ecosystem. Encourage students to recognize these parts and narrate them in turn.

★ TEACHER NOTE
Geologic land formation is referred to on p.19: "...during the last ice sheet retreat." Teachers might allow it to pass by the way or consider alternative theories they might like to share.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ 15m Natural History: Grade 6 - Lesson 16

An Island Out of Balance

Materials: The Wolves and Moose of Isle Royale

➔ RECAP
Recall what we learned in the last passage. What do you think are the differences between studying in the laboratory and studying in the field? Let's find out what's going on in this field study.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 2

→ READ, NARRATE, & DISCUSS
p.27-31 "The Isle Royale" - "National Park Service."

• When we study in the laboratory, we usually try to simplify and focus on just one thing, but the field is more complicated. What kinds of things do you see in the pictures and hear in the words of the story that make studying in the field more complicated? Can you understand why Candy Peterson thinks Isle Royale is more like a circus than a laboratory? What do you like or not like about her comparison?
• Discuss the food chain/web based on the reading OR for predators and prey in your area. Draw or diagram, if helpful.

WEEK 5

⬚ 15m Natural History: Grade 6 - Lesson 17
A Plan to Restore Balance

Materials: The Wolves and Moose of Isle Royale

PREP: Read Teacher Tip

→ RECAP
Recall what we learned in the last passage. When ideas and situations become complicated and people have different perspectives, they often disagree about what decisions are best. This happened with Isle Royale, and we'll find out more about that today.

→ READ, NARRATE, & DISCUSS
p.33-39 "What were" - "four-year-old female."

→ SUPPLEMENTAL
Scientists decided that wolf reintroduction was the best decision. Watch this PBS video link to see what happened when this keystone species was reintroduced at Yellowstone.
∞ Video Link: Wolves of Yellowstone (5:19)

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in wolves or predator-prey relationships? More interest/ease in discussing field work or tough decisions in field work? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

⬚ 15m Natural History: Grade 6 - Lesson 18
Releasing Wolves

Materials: The Wolves and Moose of Isle Royale

→ RECAP
Remind me what the problem is at Isle Royale. Recall what scientists have decided to do about this. How do you think they actually did this?

→ READ, NARRATE, & DISCUSS
p.41-47 "The team" - "in all of this."

• Explain the process of reintroducing a wolf at Isle Royale. What do you think is the most challenging part?

→ SUPPLEMENTAL
There are two wolf species featured in our book, which you can read about on p.40-41, if desired. Watch a brief video of the first female wolf being released onto the island. What do you think as you watch her?
∞ Video Link: Isle Royale Releasing Wolves (1:15)

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ 15m Natural History: Grade 6 - Lesson 19
Learning from Scat and Fur

Materials: The Wolves and Moose of Isle Royale

→ RECAP
Tell me about what you remember from the last reading. Once the wolves are released, how do you think the scientists continue to study them?

→ READ, NARRATE, & DISCUSS
p.51-58 "Morgan and I" - "to the lodge."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ 15m Natural History: Grade 6 - Lesson 20
Learning from Moose

Materials: The Wolves and Moose of Isle Royale

→ RECAP
Remind me what the last reading was about. Let's find out how scientists learn from the moose today.

→ READ, NARRATE, & DISCUSS
p.58-66 "We hike miles" - "research firsthand."

→ SUPPLEMENTAL
∞ Video Link: Outside Science (4:34)

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 2

WEEK 9	⬚ **15m Natural History: Grade 6 - Lesson 21** *Finding Moose* ▯ Materials: The Wolves and Moose of Isle Royale **→ RECAP** Tell me how the wolves and moose are studied at Isle Royale. Do you think the scientists get to watch the real-life animals? How often do you think that happens? What do you think that would be like? **→ READ, NARRATE, & DISCUSS** p.69-74 "We spot" - "us to sleep." • What are some of the activities of a field scientist while outdoors? Compare/contrast to what you do when you are on a walk. • Make up a poem about wolves together!	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 10	⬚ **15m Natural History: Grade 6 - Lesson 22** *Visiting the Circus* ▯ Materials: The Wolves and Moose of Isle Royale **→ RECAP** Tell me what you have learned about the wolves and moose at Isle Royale. Let's finish the story and find out how the ecosystem is today. **→ READ, NARRATE, & DISCUSS** p.76-82 "Let's return" - "and his DNA." • Compare Candy's perspective to man's appointed task of dominion over creation (Genesis 1:28). **→ SUPPLEMENTAL** ∞ Video Link: Isle Royale Nature Video (4:08)	★ TEACHER NOTE **Evolution is mentioned on p.79:** "... have evolved into..." Teachers might choose to let it pass by the way or discuss with students. **• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 11	⬚ **15m Natural History: Grade 6 - Lesson 23** ▯ Materials: The Wolves and Moose of Isle Royale PREP: Read Teacher Tip **→ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	★ TEACHER TIP When students take exams next week, evaluate what they recall, but more importantly, whether wolves, the role of predators in an ecosystem, or field science are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? **• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **15m Natural History: Grade 6 - Lesson 24** *Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: The Wolves and Moose of Isle Royale	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 3

WEEK 1

⬚ **15m Natural History: Grade 6 - Lesson 25**

Frogging

▯ Materials: The Frog Scientist

PREP: Read Teacher Tip.

➔ INTRO
Why do you think frogs are important? What can we learn from them? Do they impact humans? This is a story about scientists who study how frogs grow in an environment contaminated with pesticides.

➔ READ, NARRATE, & DISCUSS
p.1-3 "The sun is" - "through the mud."

• Have you ever caught or raised tadpoles? Would you like to? How do they feel in your hand, or how do you imagine they would feel in your hand?

➔ SUPPLEMENTAL
∞ Video Link: Do We Really Need Pesticides? (5:18)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, liven the lesson, or add to the discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

⬚ **15m Natural History: Grade 6 - Lesson 26**

...And Working

▯ Materials: The Frog Scientist

➔ RECAP
Recap the last reading.

➔ READ, NARRATE, & DISCUSS
p.4-7 "Forty-eight..." - "in our water?"

• Is it okay to sacrifice some animals to save others or to save human beings? Try to understand both points of view.

★ TEACHER NOTE
Scientists cull frogs, and the feminizing effect of chemicals on some frogs is described on p.6. Teachers might choose to let them pass by the way or to discuss one or both topics with students.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

⬚ **15m Natural History: Grade 6 - Lesson 27**

The Frog Scientist

▯ Materials: The Frog Scientist

➔ RECAP
Recap the last reading.

➔ READ, NARRATE, & DISCUSS
p.8-11 "Tyrone didn't worry" - "frogs were dying."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ **15m Natural History: Grade 6 - Lesson 28**

Fragile Frogs

▯ Materials: The Frog Scientist

➔ RECAP
Recap the last reading.

➔ READ, NARRATE, & DISCUSS
p.12-17 "Amphibian scientists" - "world's amphibians?"

➔ SUPPLEMENTAL
∞ Resource Link: Keeping Tadpoles

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

⬚ **15m Natural History: Grade 6 - Lesson 29**

Pesticides

▯ Materials: The Frog Scientist

PREP: Read Teacher Tip

➔ RECAP
Recall the last reading.

➔ READ, NARRATE, & DISCUSS
p.17-19 "The answer" - "to atrazine."

• Consider alternatives to pesticides. Maybe check out this video:
∞ Video Link: Hungry Ducks (2:30)

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in frogs, their life cycle, or their sensitivity to ecological factors? More interest/ease in discussing field work or lab experiments? If not, consider observing together more often, exploring extra helpings, or taking a field trip to a nature preserve or natural history museum.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 3

WEEK 6

⬚ **15m Natural History: Grade 6 - Lesson 30**
The Amphibian Ark

Materials: The Frog Scientist

→ RECAP
Recap the last reading.

→ READ, NARRATE, & DISCUSS
p.20-21 "In June 2005," - "Jump in!"

→ SUPPLEMENTAL
Have you ever seen frogs from other places in the world? If so, recall one. If not, explore this link:
∞ Resource Link: Amphibia Web

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ **15m Natural History: Grade 6 - Lesson 31**
From Lab to Field

Materials: The Frog Scientist

→ RECAP
Recap the last reading.

→ READ, NARRATE, & DISCUSS
p.22-27 "Young remembers her" - "harming frogs."

• If you have raised frogs, describe how they develop from egg to frog. If not, check out this link:
∞ Video Link: Raising frogs - Frog development (6:57)

★ TEACHER NOTE
The problem of pesticides feminizing the frogs is further described on p.26-27. Teachers might choose to discuss further with students.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ **15m Natural History: Grade 6 - Lesson 32**
The Question

Materials: The Frog Scientist

PREP: Read Teacher Tip.

→ RECAP
Recall the last reading.

→ READ, NARRATE, & DISCUSS
p.28-33 "It's early on" - "blue-yellow-yellow jug."

• Discuss the design of the experiment. What question are the scientists asking? How are they making sure the test is unbiased?

★ TEACHER TIP
The traditional names for the manipulated and responding variables are independent and dependent. Be sure students understand how they function in the experiment.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 9

⬚ **15m Natural History: Grade 6 - Lesson 33**
The Experiment

Materials: The Frog Scientist

→ RECAP
Recap the last reading.

→ READ, NARRATE, & DISCUSS
p.33-36 "Jasmin picks up" - "a wicked laugh."

• Notice the diagram of Dr. Hayes's experimental design on p.37. Discuss what makes this an excellent test of his hypothesis.

★ TEACHER NOTE
The group's Fifth of July celebration (rather than Fourth of July) is mentioned on p.36. Teachers might choose to discuss further with students.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10

⬚ **15m Natural History: Grade 6 - Lesson 34**
The Answer

Materials: The Frog Scientist

→ RECAP
Recap the last reading.

→ READ, NARRATE, & DISCUSS
p.38-45 "Tyrone's main lab" - "feminized frogs."

★ TEACHER NOTE
A photograph of preserved frog specimens is shown on p.39. Teachers might choose to cover the image or discuss beforehand with students if concerned they will be disturbed.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11

⬚ **15m Natural History: Grade 6 - Lesson 35**
Values

Materials: The Frog Scientist

PREP: Read Teacher Tip

★ TEACHER NOTE
An interaction between the scientist and his father is considered on p.46-51. Teachers might choose to preread, to use the discussion prompt, or to skip this final lesson.

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Term 3

	→ RECAP Recap the last reading. **→ READ, NARRATE, & DISCUSS** p.46-51 "When Tyrone" - " and the same." • It's very important that we think about our choices. Are they consistent with our values? What are the consequences of our choices? It's also important to know that we'll sometimes make mistakes, and we have to work to correct those mistakes. How do we see this in Dr. Hayes's work?	★ TEACHER TIP When students take exams next week, evaluate what they recall, but more importantly, whether frogs, their life cycle, or their sensitivity to ecological factors are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? **• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **15m Natural History: Grade 6 - Lesson 36** *Exam Week* **→ EXAMS** Answer question(s) related to course. Questions will come from: The Frog Scientist	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Examination

Term 1 · Tracking Trash: Draw or diagram 3 different factors that help create ocean currents; OR explain how ocean currents affect climate and wildlife.

Term 2 · The Wolves and Moose of Isle Royale: Using words or pictures, explain how wolves affect the ecosystem; OR describe 3 ways that scientists study animals in the field.

Term 3 · The Frog Scientist: Diagram the life cycle of a frog; OR give 2 examples of chemical, biological, or environmental conditions that may impact amphibians in the wild.